

Monsters



Find the starter at:

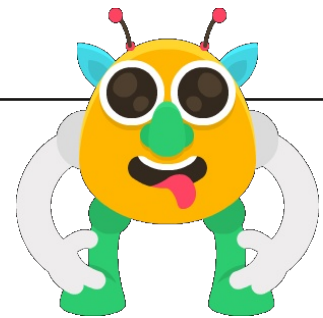
https://jrmhaig.github.io/Code_Club_Resources/python

Works best on a locally installed editor, such as **Thonny**.

1

Build a monster like Gribble, on the right. In the starter pack you will find;

- The Python script **monster.py** for you to edit.
- The **images** directory containing all the different part types.



2

Open the **monster.py** file in your editor and you will see the code on the right. This contains;

- An **import** to say we are going to use the **pgzrun** package. This is short for 'PyGame Zero Run' and it is what allows us to draw.
- Two constants **WIDTH** and **HEIGHT** that tell pgzrun how big to make the screen.
- The command **pgzrun.go()** to tell pgzrun to start. This must be at the end,

```
# This line must stay at the top
import pgzrun

WIDTH = 800
HEIGHT = 500

# This line must stay at the bottom
pgzrun.go()
```

Note: A **constant** is like a variable but it never changes for the whole program. That is why it is called a constant! A constant is normally all capitals.

after everything else.

This program will run as it is. Try it and see what happens. See what happens if you change the **WIDTH** and **HEIGHT**.

3

Something to try:

- Try changing the **WIDTH** and **HEIGHT**. Make them the right screen size for your computer.

4

pgzrun expects there to be two functions. First is **draw** which puts all the images on the screen. Add after the HEIGHT line;

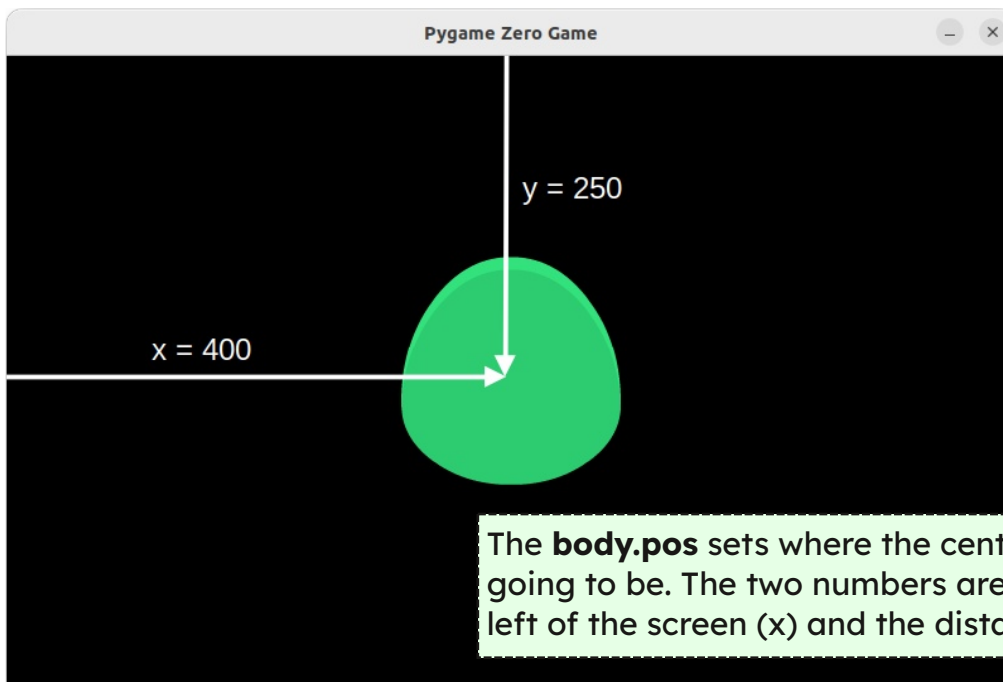
- Create a new variable **body** and set it to an Actor choosing one of the images from the bodies directory.
- Set the position of the **body** variable to the middle of the screen.
- Create the **draw** function, which will clear the screen and draw the **body**.

```
HEIGHT = 500
```

```
body = Actor('bodies/yellow_d.png')  
body.pos = (400, 250)
```

```
def draw():  
    screen.clear()  
    body.draw()
```

```
# This line must stay at the bottom
```



The **body.pos** sets where the centre of the image is going to be. The two numbers are the distance from the left of the screen (x) and the distance from the top (y).

Important note!

- Python uses **indentation** to show which code belongs inside a function or if statement.
- The spaces at the start of lines are important!

These two lines are in the **draw** function. They both have exactly **four** spaces at the start.

This line is **outside** of the draw function. It has **no** spaces at the start.

```
import pgzrun
```

```
WIDTH = 800  
HEIGHT = 500
```

```
body = Actor('bodies/yellow_d.png')  
body.pos = (400, 250)
```

```
def draw():  
    screen.clear()  
    body.draw()
```

```
pgzrun.go()
```

5

The other function pgzrun wants is **update** and it controls how things move. We will use it to control the monster with the arrow keys. After the draw function add the update function;

- Test if the left key is pressed.
- Change the x position of the body to make it move left.
- Do the same to move right.

```
def draw():
    screen.clear()
    body.draw()

def update():
    if keyboard[keys.LEFT]:
        body.x -= 5
    if keyboard[keys.RIGHT]:
        body.x += 5
    # Can you move up and down too?
```

Notice that the **if** is indented by **four** spaces and the next line is indented by four more, making **eight** in total. In Scratch it would look like this.

**6**

A couple of **optional** things you can try:

- How can you make the monster move at a different speeds?
- **(Tricky)** Can you stop the monster from going off the side of the screen?

Troubleshooting tips

If your program does not run here are some things to check.

- Notice that some of the command have brackets `()` at the end even if there is nothing in them. For example `screen.clear()` and `body.draw()`.
- The lines at the start of a block, like an `if` or a `def`, must have a colon `:` at the end. This is how Python knows where the block starts.
- The file names for the images must be exactly right. Check that you are not using capitals when they should be lower case, or if you have added extra spaces.
- **Indentation!** This is one of the hardest things for someone starting with Python. Make sure every line has exactly the right number of spaces at the start.

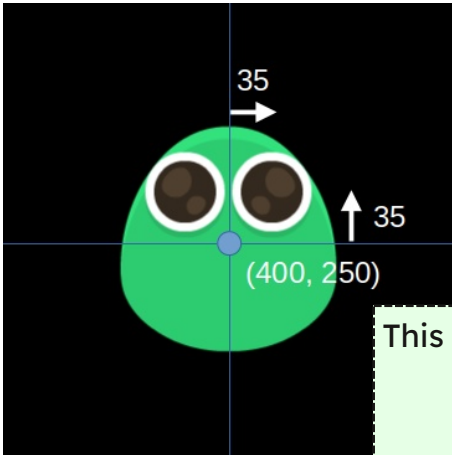
7

Now to add eyes to our monster. Changes need to be made in three places;

- At the top, add new variables for left eye and right eye.
- In the draw function, draw the two eyes.
- In the update function, set the position of the eyes relative to the body.

```
body = Actor('bodies/yellow_d.png')
body.pos = (400, 250)
leftEye = Actor('eyes/left_cute_light.png')
rightEye = Actor('eyes/right_cute_light.png')
```

```
def draw():
    screen.clear()
    body.draw()
    leftEye.draw()
    rightEye.draw()
```



```
def update():
    if keyboard[keys.LEFT]:
        body.x -= 5
    if keyboard[keys.RIGHT]:
        body.x += 5
    leftEye.pos = (body.x - 35, body.y - 35)
    rightEye.pos = (body.x + 35, body.y - 35)
```

This picture shows how we get the position of the right eye.

- The x position is the x position of the body plus 35, to place it a little to the right.
- The y position is the y position of the body minus 35, to place it a little up.

9

Some things to try:

- Can you make the eyes further apart?
- How would you make a one-eyed monster? Or a three-eyed monster?
- **(Hard)** Can you make the eyes spin?

Hint: `leftEye.angle` is like the direction of a sprite in Scratch.

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Now you can add all the other parts of the body. Browse the images directory to see what you can find. As well as more bodies and eyes, there are;

- eyebrows, noses, ears and mouths
- arms and legs
- horns

